MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER

Discipline Specific Core Course (DSC)-MG3CCRBCA201- Database Management Systems

(2024 ADMISSION ONWARDS)

Duration : 2 Hours **Maximum Marks : 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create ©, Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer All Questions

Each question carries **2** marks

1. Can a primary key be NULL or duplicate? Justify. [U][1]
2. Explain ER diagram and its use. [U][2]
3. What does the ROLLBACK command do in a transaction? [U][3]
4. Define Big Data. [U][4]
5. Describe a Foreign Key. [U][1]

(2 x 5 = 10 marks)

Part B

Short Answer Questions

Answer any **five** questions

Each question carries **6** marks

6. Analyse how constraints in a relational schema maintain integrity. [An][1]
7. Consider the table *Student(RollNo, Name, Age, Marks)*. Add a new column percentage Write SQL queries
   a) To find students whose names start with 'A'.
   b) To find students whose age is between 18 and 22.
   c) To find minimum and maximum marks of students. [A][2]
8. Illustrate the effect of using LEFT OUTER JOIN and RIGHT OUTER JOIN between two tables, orders and customers. [U][2]
9. What is the purpose of normalization in database design.? [U][3]
10. Differentiate between partial and fully functional dependencies with example. [An][3]
11. Differentiate between conflict serializability and view serializability. [U][3]

12. Write about update operators in MongoDB. [U][4]

(5 x 6 =30 Marks)

Part C

Essay Questions

Answer any **2** questions.

Each question carries **15** marks.

13. Explain the notations of ER diagram.Create an Entity-Relationship (ER) diagram for a university database that includes entities like 'student', 'course', and 'instructor'. Define suitable relationships and attributes for each entity. [A][2]
14. Discuss the role of functional dependencies in determining keys, superkeys, and candidate keys. [An][3]
15. Write a note on the various data models of NoSQL databases. [U][4]

(2 x 15=30 Marks)

MAHATMA GANDHI UNIVERSITY

BCA (HONOURS) THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG3CCRBCA202 -Software Engineering

Model Question Paper

Max. Marks: 50 **Time 1.5 Hours**

Part A: Very Short Answer Questions (Answer all) - (10*2=20 Marks)

1	Distinguish between plan-driven and agile approaches.	[U][CO 1]
2	Describe incremental delivery in agile development?	[U][CO 1]
3	Describe the key activities in the Extreme Programming (XP) process.	[U][CO 1]
4	What are the agile development tools.	[U][CO 1]
5	List the characteristics of plan-driven development.	[R][CO 2]
6	Describe project planning in software engineering?	[U][CO 2]
7	What is the purpose of a Software Requirements Document?	[An][CO 2]
8	Define cohesion as a design metric.	[R][CO 3]
9	What are the objectives of software testing?	[U][CO 3]
10	Analyze advantages and limitations of white-box testing.	[An][CO 3]

(10*2=20 Marks)

Part B: Short Answer Questions (3 out of 5 Questions)- (3*5=15 Marks)

11	Compare Scrum and Extreme Programming (XP).	[An][CO 1]
12	Explain the significance of methods and tools in software engineering.	[U][CO 1]
13	Distinguish framework activities and umbrella activities. 297328	[An][CO 1]
14	Explain experience-based techniques in software effort and cost estimation.	[U][CO 2]
15	What is the significance of design model metrics?	[U][CO 3]

(3*5=15 Marks)

Part C: Essay Questions (1 out of 2 Questions) - (1*15=15 Marks)

16	Explain requirements elicitation and analysis.	[An][CO 2]
17	Explain different system testing approaches with suitable examples	[U][CO 3]

(1*15=15 Marks)

QP Code **Reg.No:.....**

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) BCA EXAMINATION OCOTOBER 2025
III SEMESTER
Course Nature - MG3CCRBCA203- DESIGN AND ANALYSIS OF ALGORITHMS
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create(C), Skill(S), Interest (I) and Appreciation (Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer ***all*** questions
Each question carries 2 marks

1. Write any four characteristics of an algorithm. [U] / [CO1]
2. Define Recursion. [K] / [CO1]
3. State the time complexity of Strassen's matrix multiplication. [U] / [CO2]
4. Write the control abstraction of Divide and Conquer method? [R] / [CO2]
5. Give any two applications of the Greedy Method. [U] / [CO2]
6. For an array of 6 elements, compare the number of comparisons required to find both the maximum and minimum using Linear Search and Divide and Conquer methods [A] / [CO2]
7. Define principle of optimality [K] / [CO3]
8. Write the recurrence relation for the all pairs shortest path problem. [U] / [CO3]
9. Define Hamiltonian circuit. [U] / [CO3]
10. What do you mean by connected components? [U] / [CO3]

[2x10 = 20]

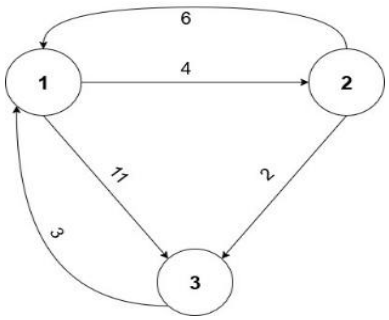
Part B
Short Answer Questions
Answer ***any three*** questions
Each question carries 5 marks

11. Explain various asymptotic notations with example. [U] / [CO1]

12. Solve a fractional knapsack problem with 3 items having weights (10, 20, 30, 25, 28) and profits (60, 100, 120, 75, 140) with capacity 50 using the greedy approach. [A] / [CO2]
13. With an example, explain how the shortest path estimate of a vertex is updated in Dijkstra's Algorithm. [U] / [CO2]

14. Explain the general concepts of dynamic programming. [U] / [CO3]

15. Apply the Floyd-Warshall algorithm to the given 3-vertex graph and determine the all-pairs shortest path matrix. [A]/[CO3]



[5x3 = 15]

Part C

Essay Questions

Answer ***any one*** question

Each question carries 15 marks

16. Construct the Minimum Spanning Tree using Prim's algorithm for the given weighted graph. Show each step clearly. Given the undirected weighted graph G with 7 vertices (A, B, C, D, E, F, G) and 15 edges E= {(A, B) = 7, (A, D) = 5, (B, C) = 8, (B, D) = 9, (B, E) = 7, (C, E) = 5, (C, F) = 6, (D, E) = 15, (D, F) = 10, (E, F) = 8, (E, G) = 9, (F, G) = 11, (A, C) = 12, (B, F) = 14, (C, G) = 13}

[A] / [CO2]

17. Explain the Backtracking approach to solve the N-Queen's problem? Write an algorithm to solve it for value of N=8. [U] / [CO3]

[15x1 = 15]

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER

Discipline Specific Core Course (DSC)- MG3DSEBCA200- Basics of Data Analytics using Spreadsheet

(2024 ADMISSION ONWARDS)

Duration : 1 Hour **Maximum Marks : 35**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create ©, Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Short Answer Questions
Answer any **seven** questions
Each question carries **five** marks

1. Define structured data with two examples [R][1]
2. Explain diagnostic analytics with example [U][1]
3. Discuss how does data analytics contribute to competitive advantage for businesses? Explain with examples. [A][1]
4. Describe how data analytics is used in e-commerce to enhance sales and customer engagement. [A][1]
5. List different methods of collecting primary data and briefly explain any two. [U][2]
6. List out the effects of noisy data in data analytics. [U][2]
7. Compare Min-Max normalization and Z-Score normalization? [U][2]
8. Determine the significance of outliers in influencing the mean and median of a dataset. [U][2]
9. Demonstrate how informed consent can be implemented during data collection? [U][2]
10. A telecom company is losing customers. Apply predictive analytics to suggest a solution to reduce this problem. [A][2]

(7x5=35 marks)

MAHATMA GANDHI UNIVERSITY

BCA (HONOURS) THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
MG3DSEBCA201 - Feature Engineering
Model Question Paper

Maximum Marks : 35 **Time: 1 Hour**

Part A : Very Short Answer Questions(Answer all) (10*2=20 Marks)

1	Differentiate between data and information.	[U][CO 1]
2	Distinguish between discrete data and continuous data with examples.	[U][CO 1]
3	What is the significance of ordinal data in decision making?	[U][CO 1]
4	What are the common methods for handling missing data?	[U][CO 2]
5	Define data cleaning.	[R][CO 2]
6	Write the formula for Min-Max normalisation	[U][CO 2]
7	Distinguish between Equal Width and Equal Frequency discretization	[An][CO 3]
8	Define data binning	[R][CO 3]
9	What are the advantages of One-Hot Encoding?	[U][CO 4]
10	What are the limitations of Label Encoding?	[An][CO 4]

(10*2=20 Marks)

Part B : Short answer questions (3 out of 5 questions) (3*5=15 Marks)

11	Differentiate between interval and ratio scales with suitable examples.	[An][CO 1]
12	Explain different types of data transformation techniques.	[U][CO 2]
13	Explain how the interaction feature affects a model's prediction.	[An][CO 3]
14	Distinguish between feature extraction and feature selection with suitable examples.	[An][CO 4]
15	Explain how Recursive Feature Elimination (RFE) works as a wrapper method for feature selection.	[An][CO 4]

(3*5=15 Marks)

QPCode **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION
THIRD SEMESTER
MG3DSEBCA202 WEB PROGRAMMING - I
(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer all questions. Each question carries 2 marks

1.	List any four commonly used web browsers.	K /CO 1
2.	List two rules for naming variables in PHP.	K /CO 1
3.	Write a PHP script to print "Hello World" using echo	A /CO 1
4.	Mention the special datatypes of PHP.	U /CO 1
5.	Differentiate between == and === used in PHP.	A /CO 1
6.	What is the purpose of associative arrays in PHP?	U /CO 2
7.	What does the array_unshift() function do in PHP?	U /CO 2
8.	What is the purpose of the $_GET super global in PHP?	K /CO 2
9.	What is the purpose of the include function in PHP?	K /CO 3
10.	Write the syntax of mysqli_connect() function.	A /CO 3

[10 x 2 = 20]

Part B

Short Answer Questions

Answer any 3 questions. Each question carries 5 marks

11.	Discuss the difference between client-side and server-side scripting.	U /CO 1
12.	Illustrate all the decision control statements of PHP with an example	A /CO 2
13.	Explain foreach loop used in PHP with an Example	U /CO 2
14.	Write a PHP script to perform the operations Delete and Select on a students table.	A /CO 3
15.	Explain mysqli_fetch_row(), mysqli_fetch_array(), and mysqli_fetch_assoc() with suitable examples.	U /CO 3

[3 x 5 = 15]

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU BCA (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3SECBCA200: PYTHON PROGRAMMING

MODEL QUESTION PAPER

Duration: 1.5 Hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A: Very Short Answer Questions.

Answer All Questions. Each question carries 2 marks.

1. Demonstrate the significance of indentation in Python programming. [U][1]
2. Discuss the difference between `=` and `==` operators in Python. [U][1]
3. Write a Python statement to extract "gram" from "Programming". [U][1]
4. State the functions of break and continue statement in Python. [K][1]
5. Write a user-defined function to find largest of two numbers. [A][2]
6. Show with an example how to add a key-value pair to a dictionary. [A][2]
7. State the difference between `import module` and `from module import function`. [K][2]
8. Write a Python code using Numpy to generate an array of even numbers between 10 and 20. [A][2]
9. Define a text file and a binary file. [A][3]
10. State the purpose of the try block in Python? [K][3]

(10 X 2 = 20 Marks)

Part B: Short Answer Questions.

Answer any 3 out of 5 Questions. Each question carries 5 marks

11. Construct a python program that checks whether a given year is leap year or not.[A][1]
12. Write a program that uses the `assert` statement to check if a given number is positive, and display an error message if it is not. [A][2]

13. Differentiate between `pop()`, `popitem()`, and `clear()` functions in dictionaries with suitable illustrations. [U][2]
14. Differentiate between stacked bar chart and grouped bar chart. [U][3]
15. Compare 2D and 3D plotting with examples. [U][3]

(3X 5 = 15 Marks)

Part C: Essay Questions.

Answer any 1 out of 2 Questions. Each question carries 15 marks

16. Explain the role of `append()`, `extend()`, `insert()`, `remove()`, `pop()`, `clear()`, and `sort()` in lists with proper code illustrations. [U][2]
17. Outline different loop control statements in Python with suitable examples. [U][1]

(1 X 15 = 15 Marks)